

ANUSTUP BHAUMIK

MSc in Computer Science, RKMVERI, Belur

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SUMMARY

I am a curious learner with a strong background in algorithms, machine learning, and data science. My experience spans designing and optimizing deep learning architectures, analyzing complex datasets, and building high-performance models. I combine a solid theoretical foundation with a practical, engineering-first approach to solve complex data challenges. I am also a proponent of free and open source software.

EDUCATION

University of Kalyani

BSc (Honours) in Computer Science, 2022–25
CGPA –7.93

Ramakrishna Mission Vivekananda Educational and Research Institute

MSc in Computer Science, 2025–27

EXPERIENCE

NISER, Bhubaneswar

Summer Research Intern, May 2026 - July 2026
Supervisor: Dr. Aritra Banik

SKILLS

- **Languages:** Python, C, C++, HTML, Bash
- **Frameworks:** NumPy, Scikit-Learn, PyTorch, Matplotlib, OpenGL
- **Databases:** PostgreSQL, neo4j, OracleSQL
- **Tools:** Git, LaTeX, MS Office
- **Platforms:** GNU/Linux, Windows

COURSE WORK

- Large Language Models, Applied Deep Learning, Deep Learning and NLP
- Machine Learning, Artificial Intelligence, Data Mining
- Probability and Statistics, Linear Algebra
- Advanced Algorithms, Data Structures, Spectral Graph Theory, TOC

ACHIEVEMENTS

- CUET-PG 2025 (Computer Science) – Qualified
- Best Paper Award in Basic and Applied Sciences category at ICACCSFK 2025

CERTIFICATES

- **Way to AI**
3-months Add-On Course
Dept of Computer Science
Kanchrapara College
- **Ajit Diwan Memorial Workshop on Geometry, Graph & Combinatorics**
Dept of Computer Science
Ramakrishna Mission Vivekananda Educational and Research Institute

PROJECTS

LLM powered Natural Language to SQL query generator

RKMVERI

May 2026 – July 2026

Built a full-stack text-to-SQL system powered by local Llama models via Ollama. To make it smart, I added semantic retrieval using sentence-transformer embeddings. This helps the AI truly understand the database schema, grab relevant business context, and pull past query examples to write highly accurate SQL every time.

Deep Learning based Network Intrusion Detection System

RKMVERI

Feb 2026 – Apr 2026

Evaluated XGBoost, MLP, and hybrid CNN-LSTM models for intrusion detection using the 2.8M-sample CICIDS2017 dataset. Engineered PyTorch and Scikit-learn pipelines to normalize 79 features and resolve severe class imbalance via SMOTE. Restructured tabular traffic into temporal sliding windows, optimizing a 1D-CNN with dual-layer LSTMs to achieve a 99.94 % macro F1-score on multi-stage cyber attacks.

SkyRoute:Autonomous Drone Delivery System using Approximate Q-Learning

RKMVERI

Oct 2025 – Dec 2025

Developed an autonomous drone navigation system using Approximate Q-Learning with linear function approximation in a custom Gymnasium environment. Engineered feature extractors factoring in distance-to-goal metrics, obstacle maps, and steady wind fields. Implemented an exponential repulsive potential field to dynamically calculate hazard vectors and evade moving obstacles. Optimized trajectories across 3,000+ training episodes to mitigate battery drain and eliminate collision rates.

Grid-World Agent Optimization via Reinforcement Learning

RKMVERI

Oct 2025 – Dec 2025

Developed a tabular Q-learning agent within the OpenAI Gymnasium Taxi-v3 environment to optimize autonomous grid-world navigation and passenger delivery. Applied an ϵ -greedy exploration strategy alongside Bellman equation updates to iteratively converge an optimal 500×6 policy matrix. Evaluated model execution through real-time human rendering modes and imageio pipelines, successfully stabilizing decision-making to complete passenger transit in 11 steps with maximum cumulative reward.

Autism Spectrum Disorder Prediction using various Classification Models

RKMVERI

Sep 2025 – Nov 2025

Developed an autism prediction system using clinical and behavioral dataset features, including AQ-10 scores, demographics, and medical history. Pre-processed indicators and utilized SMOTE to balance the target distribution across 750 patient records. Conducted a comparative evaluation of seven classification models, optimizing hyperparameters via RandomizedSearchCV with 5-fold cross-validation. Selected Logistic Regression as the deployment baseline, maximizing clinical screening utility with a 0.87 recall and 0.85 testing accuracy while preventing overfitting.